

Lagrangian mechanics

Description of motion through energy

Notation

velocity = $\frac{dr}{dt}$ linear momenta : mv force : F

angular momentum $L : r \times p$

angular Torque $\tau = r \times F$

Newton's second law of motion $F = \frac{dp}{dt}$

angular version $\tau = r \times \frac{dp}{dt} = \frac{d(r \times p)}{dt} - \frac{dr}{dt} \times p$

$$= \frac{dL}{dt} - \underbrace{v \times mv}_0 \Rightarrow \boxed{\tau = \frac{dL}{dt}}$$

$$F = \Delta \mathcal{E}_c \Rightarrow F = -\Delta \mathcal{E}_p$$

$$F = -\nabla \mathcal{E}_p(r)$$

chegamos em outra conclusão

$$W_{12} = \frac{mv_2^2}{2} - \frac{mv_1^2}{2} = T_2 - T_1$$

$$W_{12} = E_{p1} - E_{p2} \quad E_p \Rightarrow V$$

$$\rightarrow T_2 - T_1 = V_1 - V_2$$

$$\rightarrow T_1 + V_1 = T_2 + V_2$$

$$E_{c1} + E_{p1} = E_{c2} + E_{p2}$$

} é CTD para sistemas conservativos

o Sistema de muitas partículas

Par a partícula i

$$F_i = \dot{P} = \frac{d(mv)}{dt}$$

$$F_{i(ext)} + \underbrace{\sum_j F_{ji}}_0 = \dot{P}_i$$

$$\sum_i F_{i(ext)} = \frac{d^2}{dt^2} \sum_i (m_i v_i)$$

$$\sim \sum F_{ext} = \sum \Delta P \sim \text{simplificar}$$

Definimos Centro de Massa

$$R = \frac{\sum m_i r_i}{\sum m_i} = \frac{\sum m_i r_i}{M}$$

$$\sum F_{i, \text{ext}} = \frac{d^2}{dt^2} M R$$

$$\sum F_{i, \text{ext}} = M \frac{d^2}{dt^2} R$$

• a variação da posição do centro de massa só ocorre sobre ação de força externas

Se $\sum F_{i, \text{ext}} = 0$ p is conserved
"weak law of action & reaction"

Performing the same line of reasoning for angular momentum (L)

$$L = \sum_i r_i \times p_i$$

$$\frac{dL}{dt} = \sum_i r_i \times F_{i, \text{ext}} + \underbrace{\sum_{\substack{i, j \\ i \neq j}} r_i \times F_{ji}}$$

$$r_i \times F_{ji} + r_j \times F_{ij} \rightarrow$$

$$(\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{F}_{ji}$$

$$\rightarrow \mathbf{F}_{ji} = -\mathbf{F}_{ij}$$

$$\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$$

$$\mathbf{r}_{ij} \times \mathbf{F}_{ji}$$

$$\frac{d\mathbf{L}}{dt} = \tau^{\text{ext}} + \underbrace{\frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} \mathbf{r}_{ij} \times \mathbf{F}_{ji}}_0$$

$$\underline{\underline{\frac{d\mathbf{L}}{dt} = \tau^{\text{ext}}}}$$

O vetor característico de qualquer partícula i pode ser escrito como

$$\mathbf{r}_i = \mathbf{R} + \mathbf{r}_i' \quad \text{onde } \mathbf{r}_i' \text{ é a distância da partícula ao CM} \rightarrow \mathbf{R}$$

Da mesma maneira

$$\mathbf{v}_i = \mathbf{v}_{\text{cm}} + \mathbf{v}_i'$$

$$\text{So } L = \sum_i \mathbf{r}_i \times \mathbf{p}_i \Rightarrow L = \sum (\mathbf{r}_i' + \mathbf{R}) \times (\mathbf{V} + \mathbf{v}_i') m$$

$$\sum \mathbf{R} \times m_i \mathbf{V} + \sum \mathbf{r}_i' \times m_i \mathbf{V} + \sum m_i \mathbf{r}_i' \times \mathbf{V} + \sum \mathbf{R} \times m_i \mathbf{v}_i'$$

$$\sum \mathbf{R} \times m_i \mathbf{V} + \sum \mathbf{r}_i' \times m_i \mathbf{V} + \sum m_i \mathbf{r}_i' \times \mathbf{V} + \mathbf{R} \times \frac{d(\sum m_i \mathbf{r}_i')}{dt}$$

$$\rightarrow L = \underbrace{\mathbf{R} \times M \mathbf{V}}_{\substack{\text{angular} \\ \text{momentum} \\ \text{of object} = \text{at} \\ \text{CM}}} + \underbrace{\sum \mathbf{r}_i' \times \mathbf{p}_i'}_{\substack{\text{angular momentum} \\ \text{around} \\ \text{CM}}}$$

We can do the same for Kinetic Energy (T)

$$T = \frac{1}{2} \sum (m_i v_i^2) \Rightarrow \frac{1}{2} \sum m (\mathbf{V} + \mathbf{v}_i') (\mathbf{V} + \mathbf{v}_i')$$

$$T = \frac{1}{2} \sum m V^2 + \underbrace{\sum m \mathbf{V} \cdot \mathbf{v}_i'}_0 + \frac{1}{2} \sum m_i v_i'^2$$

$$T = \underbrace{\frac{1}{2} M V^2}_{\substack{\text{Kinetic motion} \\ \text{of CM}}} + \underbrace{\frac{1}{2} \sum m_i v_i'^2}_{\substack{\text{Kinetic} \\ \text{motion around CM}}}$$

Limites e Coordenadas Generalizadas

Limites / Constraints

"partícula de um gás no container nunca pode ter r_i fora do container"

"Bola rolando em plano inclinado nunca pode ter um r_i na qual $\vec{p} \cdot \vec{t}$ tangencial > 0"

Holonomic Constraints

$$f(r_1, r_2, \dots, z) = 0$$

an example is =

Rigid Body equation $(r_i - r_j)^2 - C_{ij}^2 = 0$

a distância entre dois pontos tem que ser C_{ij}

non holonomic constraints não podem ser escritos como $x, f(r_1, r_2, \dots, z)$

um exemplo é uma pessoa subindo um hill sua posição pode ser h ou a cima do chão mas não dentro do mesmo de o raio da moeda é a

$$r^2 - a^2 \geq 0$$

Coordenadas Generalizadas

Se tivermos um sistema de N partículas em 3D

nós dizemos que no total tem $3N$ graus de liberdade

Se tem K restrições holônicas presentes
o grau de liberdade diminui

$$\lambda = 3N - K$$

Matematicamente deixe r_i denotar o vetor posição
de N partícula, no entanto nenhum desses
vetores r_i podem ser independentes já que
há limitações impostas no sistema

Por exemplo no caso de um rigid body
a distância entre duas quaisquer partículas é fixa

assim há $3N - K$ coordenadas independentes
que chamamos de $q_1, q_2, q_3, \dots, q_{3N-K}$
estas coordenadas generalizadas

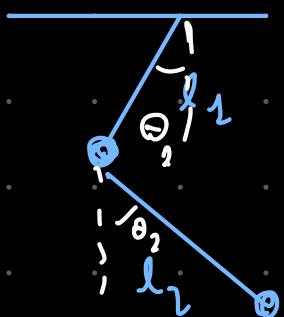
têm em conta as limitações impostas no sistema
e podem ser usadas para descrever as ^{vetores} posições
 r_i

$$r_1 = r_1(q_1, q_2, q_3, \dots, q_{3N-K})$$

$$r_2 = r_2(q_1, q_2, q_3, \dots, q_{3N-K})$$

$$r_N = r_N(q_1, q_2, q_3, \dots, q_{3N-K})$$

Exemplo Pendulo Duplo se movendo em Plano



$$N=2$$

$$6 - h$$

$h \rightarrow$ Plano 2 partículas = 2
 $\rightarrow$ Distância fixa 2 partículas = 2

$$h = 4$$

$$\lambda = 2$$

2 graus de liberdade e assim duas coordenadas generalizadas seriam essas

$$\theta_1 \text{ e } \theta_2$$

Princípio de D'Alembert e Equações de Lagrange define-se δr_i como o deslocamento infinitesimal de coordenadas do sistema com respeito aos limites em pontos

Para um sistema em eq $F_i = 0$

$$\sum F_i \delta r_i = 0$$

agora as forças aplicadas na partícula F_i podem ser separada em F_i^a e f_i

$$F_i = F_i^a + f_i$$

forças
externamente
aplicadas $\quad$ forças
dos constraints

é razoável assumir que $f_i \delta r_i = 0$

Se que todas forças por constraint são
exercidas normalmente $\Rightarrow$ deslocamento ex
Normal

$$\sum F_i^a \delta r_i + \sum f_i \delta r_i = 0$$

$$\sum F_i^a \delta r_i = 0$$

Para um sistema fora do equilíbrio

2nd lei $F_i - \dot{p} = 0$

$$F_i^a + f_i - \dot{p} = 0$$

$$\sum (F_i^a + f_i - \dot{p}) \delta r_i = 0$$

$$\sum (F_i^a - \dot{p}) \delta r_i = 0 \quad] \text{ generalizado}$$

$$\sum_i (F_i^a - \dot{p}_i) \delta r_i = 0 \quad \text{Princípio de D'Alembert}$$

Supondo

holonomic constraints

reintroduzindo as coordenadas generalizadas

$$r_i = r_i(q_1, q_2, \dots, q_n, t)$$

Temos que a velocidade da partícula
[isto é o vetor de velocidade]

$$\vec{v}_i \Rightarrow \vec{v}_i(q_1, q_2, \tau)$$

$$\frac{d}{d\tau} r_i = \frac{\partial r_i}{\partial q_i} \frac{dq_i}{d\tau} + \frac{\partial r_i}{\partial \tau} \frac{d\tau}{d\tau}$$

$$v_i = \frac{\partial r_i}{\partial q_i} \dot{q}_i + \frac{\partial r_i}{\partial \tau} \Rightarrow v_i(q_i, \dot{q}_i, \tau)$$

$$\vec{v}_i = \sum \frac{\partial \vec{r}_i}{\partial q_i} \dot{q}_i$$

euler lagrange equations

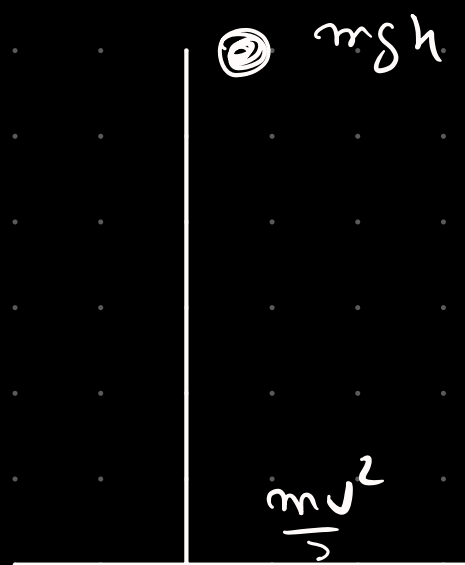
$$\bullet \quad mgh$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x}$$

$$\dot{x} = \frac{dx}{dt} = v$$

$$\frac{d}{dt} \left(m \frac{\partial \frac{v^2}{2} - gh}{\partial v} \right) = \frac{\partial [m \frac{v^2}{2} - gh]}{\partial x}$$

$$\frac{d}{dt} \left(m \frac{\partial \frac{v^2}{2}}{\partial v} \right) = -mg$$



$$\frac{m}{2} \frac{d}{dt} \frac{d}{dt} v = -mg \Rightarrow m a = -mg$$

$$\underline{a = -g}$$

Proving Energy, Momentum Conservation

• The laws of physics independent of place

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x}$$

$$\frac{\partial L}{\partial \dot{x}} \Rightarrow \frac{m v}{z} = P$$

↓
Independs from
space so

$$\frac{\partial L}{\partial x} = 0$$

$$\frac{\partial P}{\partial z} = 0$$

→ in absence of external potential

→ momentum is conserved

• laws of physics independ from time

Sim can pulci 6 pmc, an man e difinl

$$\rightarrow \text{gradient } \nabla A = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \frac{\partial A_i}{\partial x_i}$$

$$\partial_i A_i = \nabla A$$

$$\nabla \phi = e_i \partial_i \phi$$

gradient
as a
comp
escale

↓
basis

Gradient Theory

gradient is a vector
 rate of change which
 its components are the partial derivatives
 of a function

a function is $f(x, y)$

it can change on x and y $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

Gradient dot and cross product

the gradient here acts like a operator

→ Divergence $\text{div } \psi = \nabla \cdot \psi = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \cdot (\psi_x, \psi_y)$

$$\nabla \cdot \psi = \left(\frac{\partial \psi_x}{\partial x}, \frac{\partial \psi_y}{\partial y} \right)$$

→ Curl

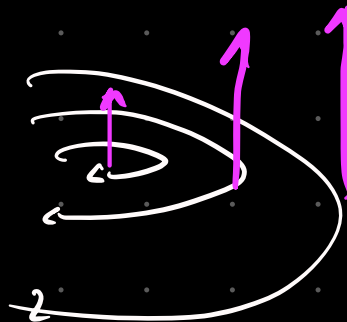
$$\text{Curl } \psi = \nabla \times \psi = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \times (\psi_x, \psi_y)$$

$$\nabla \times \psi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ \psi_x & \psi_y & 0 \end{vmatrix}$$

$$\hat{k} \frac{\partial \psi_y}{\partial x} - \hat{i} \left(\frac{\partial \psi_x}{\partial y} \right)$$

$$\nabla \times \psi = \hat{n} \left(\frac{\partial \psi_y}{\partial x} - \frac{\partial \psi_x}{\partial y} \right)$$

Curl is the
of circulation



→ Laplacian Δ

$$\Delta \psi = \nabla^2 \psi = (\nabla \cdot \nabla) \psi$$

is just $\left(\frac{\partial^2 \psi}{\partial x^2}, \frac{\partial^2 \psi}{\partial y^2}, \frac{\partial^2 \psi}{\partial z^2} \right)$

How much a function changes on a small sphere centered at a point

The Differential Maxwell Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

↓
Density
of
electric field

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$$

↙
Density of magnetic field
is dependent on the density current

is dependent on the density of magnetic field

The gradient of a function is a vector
 By itself it acts as a vector operator

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \frac{\partial}{\partial i} = \partial_i$$

$$\nabla \psi = \left(\frac{\partial \psi}{\partial x}, \frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial z} \right) = \partial_i \psi_i$$

THIS IS NOT
 the Einstein summation
 convention, it is
a vector

no livro de GR de
 SCHUTZ

ele fala que o gradient de uma função
 é um tensor (1) $\nabla \rightarrow$ não é um

$\nabla \psi$ $\rightarrow$ é tensor

então aplicar $\nabla \psi(\vec{v}) =$ retorna
 um número
 real

ele pode ser um tensor

nem sempre é interessante trata-lo como
 tal

→ Levy Civita

ϵ_{ijk} $\left\{ \begin{array}{l} \text{is } ijk \text{ permutation of } 123 \text{ then } +1 \\ \text{is } ijk \text{ permutation of } 321 \text{ then } -1 \end{array} \right.$

Optimal definition

$$\left[\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl} \right]$$

$j=l \text{ and } k=m$ $j=m \text{ and } k=l$

The Lagrangian with velocity dependent Potentials

every Lagrangian equation $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) = \frac{\partial \mathcal{L}}{\partial q_j}$

1D: $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x}$

2D: $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} + \frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = \frac{\partial \mathcal{L}}{\partial x} + \frac{\partial \mathcal{L}}{\partial y}$

Electromag

$$F = q(\vec{E} + \vec{v} \times \vec{B}) \quad \begin{array}{l} \vec{\nabla} \cdot \vec{B} = 0 \\ \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{array}$$

$$\rightarrow \vec{B} = \nabla \vec{A}$$

$$\nabla \times \vec{E} = - \frac{\partial \nabla \vec{A}}{\partial \tau}$$

$$\nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial \tau} \right) = 0$$

$$\vec{E} + \frac{\partial \vec{A}}{\partial \tau} = -\nabla \phi$$

$$\vec{F} = q \left(-\nabla \phi - \frac{\partial \vec{A}}{\partial \tau} + \vec{v} \times \nabla \times \vec{A} \right)$$

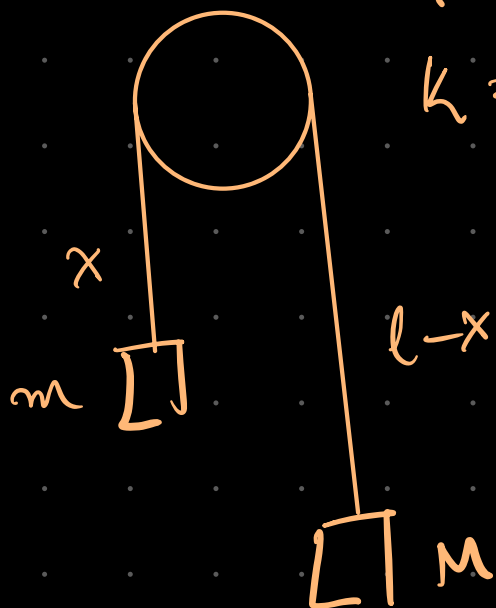
$$[\vec{v} \times (\nabla \times \vec{A})]_i = v_j \times \partial_i A_j - v_j \partial_j A_i$$

$$F_i = -\partial_i U + \frac{d}{d\tau} \frac{\partial U}{\partial v_i}$$

$$\therefore \vec{F} = -\frac{\partial U}{\partial \vec{x}} + \frac{d}{d\tau} \left(\frac{\partial U}{\partial \dot{\vec{x}}} \right) \quad U = q\phi - q\vec{A} \cdot \vec{v}$$

$$\mathcal{L} = T - U$$

$$\mathcal{L} = T - q\phi - q\vec{A} \cdot \vec{v}$$



$$T = -m_1 g x - m_2 g (l-x)$$

$$K = \frac{\dot{x}^2}{2} (m_1 + m_2)$$

$$L = \frac{\dot{x}^2}{2} (m_1 + m_2) + m_1 g x + m_2 g (l-x)$$

$$\frac{d}{dt} \int' L = \nabla L$$

$$\nabla L = m_1 g - m_2 g$$

$$\int' L = (m_1 + m_2) v$$

$$(m_1 + m_2) v = m_1 g - m_2 g$$

Equation of motion

Hamiltonian

$$H = K + T$$

$$\left[\dot{x} = \frac{\partial H}{\partial p} \quad p = - \frac{\partial H}{\partial x} \right] \text{ Hamilton's equations}$$

$$\int \partial p \dot{x} = \int \partial H$$

$$\int m \dot{x} \frac{\partial x}{\partial t} = \int \partial H$$
$$\int m \dot{x} \cdot dx = \int \partial H$$

$$\underline{W = m \dot{x} dx}$$

2

Coordenadas Generalizadas

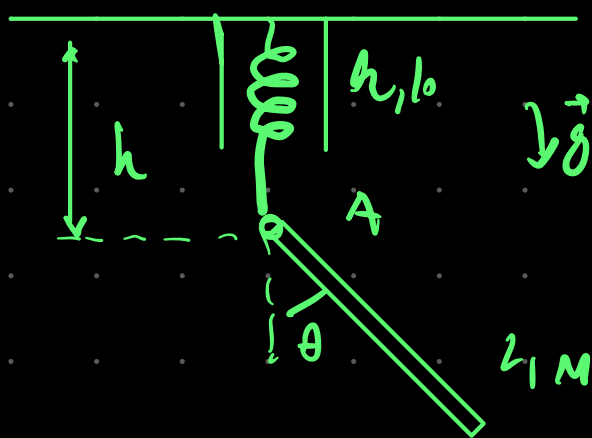
$\vec{r} = (x, y, z)$ para descrever um sistema de N partículas precisará $3N$ coordenadas

• Grau de liberdade $\equiv$ número de variáveis independentes que é necessário para conhecer completamente a posição de um sistema

• Se um partícula se move na superfície de uma esfera $\rightarrow$ 2 graus em sobre plano xy - 2 graus sobre uma linha z grau

Para 5 graus de liberdade não necessário 5 variáveis chamadas coordenadas Generalizadas (parametrizar)

Ex



A se impulsiona só em y e a barra tem a liberdade então são 2 variáveis

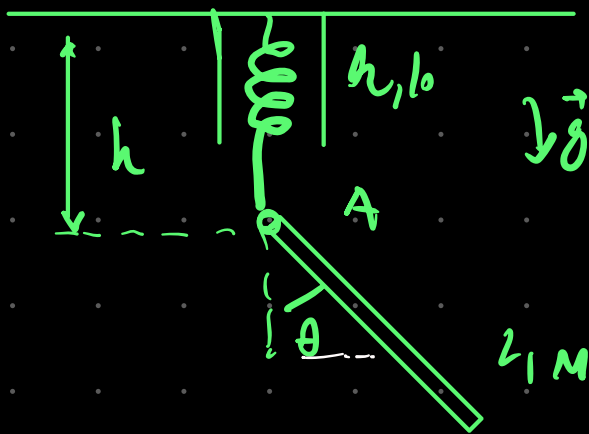
Parametrizando são h, θ

as coordenadas são definidas como $q_1, q_2, q_3, \dots, q_s$
(7)

Lagrangiano de sistema de partículas

$$\mathcal{L}(q, \dot{q}, \tau)$$

$$\mathcal{L} = T - U$$



$$T_{\text{trans, cm}} = \frac{1}{2} M V_{\text{cm}}^2$$

$$\vec{r}_{\text{cm}} = \left(h + \frac{L}{2} \cos \theta\right) \hat{i} + \left(\frac{L}{2} \sin \theta\right) \hat{j}$$

$$\vec{v}_{\text{cm}} = \frac{d\vec{r}_{\text{cm}}}{d\tau} =$$

$$= \left[\dot{h} - \frac{L}{2} \dot{\theta} \sin \theta\right] \hat{i} + \frac{L}{2} \dot{\theta} \cos \theta \hat{j}$$

$$v_{\text{cm}}^2 = \dot{h}^2 - L \dot{\theta} \dot{h} \sin \theta + \frac{L^2}{4} \dot{\theta}^2$$

$$T_{\text{Trans, cm}} = \frac{1}{2} M \left[\dot{h}^2 - L \dot{\theta} \dot{h} \sin \theta + \frac{L^2}{4} \dot{\theta}^2 \right]$$

$$T_{\text{rot, cm}} = \frac{1}{2} I_{\text{cm}} \dot{\theta}^2 = \frac{ML^2 \dot{\theta}^2}{24}$$

$$T = \frac{1}{2} M \dot{h}^2 - \frac{1}{2} M L \dot{\theta} \dot{h} \sin \theta + \frac{1}{2} M L^2 \dot{\theta}^2$$

$$K = -Mg(h - \frac{L}{2} \cos \theta) + \frac{1}{2} k (h - l_0)^2$$

$$\mathcal{L} = \frac{1}{2} M \dot{h}^2 - \frac{1}{2} M L \dot{\theta} \dot{h} \sin \theta + \frac{1}{2} M L^2 \dot{\theta}^2 + Mg(h - \frac{L}{2} \cos \theta) - \frac{1}{2} k (h - l_0)^2$$

Principle of least action

$q(\tau)$ entre τ_1 e τ_2

$$q(\tau_1) \text{ e } q(\tau_2) \quad S = \int_{\tau_1}^{\tau_2} \mathcal{L} d\tau$$

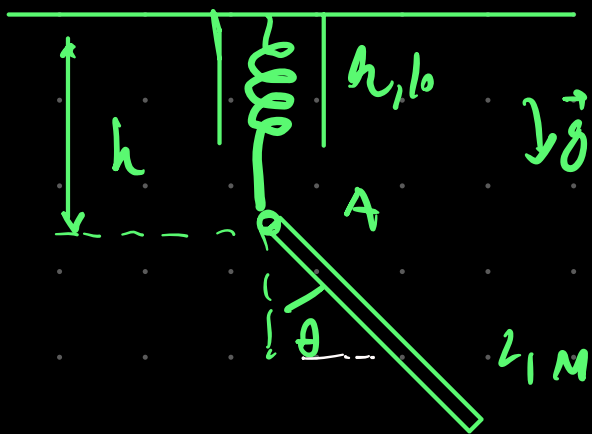
O menor valor de S é a curva

Euler-Lagrange equation

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) - \frac{\partial \mathcal{L}}{\partial q^i} = 0 \quad i = 1, 2, 3, \dots, f$$

Retorna um sistema de f equações diferenciais

Resolver essas equações retorna $q(t)$



$$\mathcal{L} = \frac{1}{2} M \dot{h}^2 - \frac{1}{2} M L \dot{\theta} \dot{h} \sin \theta + \frac{1}{6} M L^2 \dot{\theta}^2 + M g (h + \frac{L}{2} \cos \theta) - \frac{1}{2} k (h - l_0)^2$$

Per h

$$\frac{\partial \mathcal{L}}{\partial h} = M g - k (h - l_0)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{h}} = M \dot{h} - \frac{M L}{2} \dot{\theta} \sin \theta$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{h}} = M \ddot{h} - \frac{M L}{2} \ddot{\theta} \sin \theta$$

$$M \ddot{h} - \frac{M L}{2} \ddot{\theta} \sin \theta - \frac{M L}{2} \dot{\theta}^2 \cos \theta = M g - k (h - l_0) - \frac{M L}{2} \dot{\theta}^2 \cos \theta$$

$$M \ddot{h} = \frac{M L}{2} \ddot{\theta} \sin \theta + \frac{M L}{2} \dot{\theta}^2 \cos \theta + M g - k (h - l_0) \quad \textcircled{I}$$

Per θ :

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\frac{M L}{2} \dot{h} \cos \theta - \frac{M g L}{2} \sin \theta$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{M L^2}{3} \dot{\theta} - \frac{M L}{2} \dot{h} \sin \theta$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{M L^2}{3} \ddot{\theta} - \frac{M L}{2} \ddot{h} \sin \theta - \frac{M L}{2} \dot{h} \dot{\theta} \cos \theta$$

$$\frac{M L^2}{3} \ddot{\theta} = \frac{M L}{2} \ddot{h} \sin \theta - \frac{M g L}{2} \sin \theta \quad \textcircled{II}$$

$$\left\{ \begin{aligned} M \ddot{h} &= \frac{ML\ddot{\theta} \sin \theta + \frac{ML}{2} \dot{\theta}^2 \cos \theta + Mg - k(h-l_0)}{\quad} \quad (I) \end{aligned} \right.$$

$$\frac{ML^2}{3} \ddot{\theta} = \frac{ML}{2} \ddot{h} \sin(\theta) - \frac{MgL}{2} \sin \theta \quad (II)$$

$$h(\tau) \text{ e } \theta(\tau)$$

Finding the least action

$$S = \int_{t_i}^{t_f} \mathcal{L} d\tau$$

The minimum of a function is
 $f'(x) = 0$ $\Delta f(x + \epsilon) = 0$

To find the minimum function or curve
the curve $x(\tau)$ for which S is min.

$$x(\tau) \rightarrow x(\tau) + \epsilon(\tau)$$



lets determine how $\mathcal{L}$ changes

$$\mathcal{L}\left(\frac{dx}{d\tau}, x\right) = \frac{m}{2} \left(\frac{dx}{d\tau}\right)^2 + U(x)$$